

CAMBRIDGE BAY, NU, Canada

WMO: 712880

Lat: 69.107N

Lon: 105.116W

Elev: 61

StdP: 14.66

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.3	-40.7	-49.6	0.3	-42.5	-47.8	0.3	-40.6	35.5	-15.6	31.5	-17.9	8.0	40	1.106

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	63.7	55.2	59.9	52.9	56.4	50.9	55.8	62.5	53.4	59.0	51.2	55.9	11.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.0	57.7	58.2	49.8	53.2	56.3	47.7	49.1	54.1	23.7	62.7	22.3	59.2	21.0	56.3	64.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.1	27.5	24.0	DB	-45.8	70.2	3.6	4.0	-48.4	73.1	-50.5	75.4	-52.6	77.6	-55.2	80.5
			WB	-45.6	60.4	3.2	2.9	-47.9	62.5	-49.8	64.2	-51.6	65.8	-53.9	67.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-23.6	-25.3	-19.5	-4.6	16.1	37.3	48.7	45.4	33.4	15.4	-5.0	-17.8	
	DBStd	28.23	10.12	9.79	11.13	10.56	10.01	7.03	5.64	5.73	6.66	10.26	10.98	10.78	
	HDD50	15203	2283	2110	2154	1637	1050	385	96	166	497	1074	1650	2101	
	HDD65	20595	2748	2530	2619	2087	1515	830	506	608	947	1539	2100	2566	
	CDD50	83	0	0	0	0	0	5	55	23	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	12.0	11.9	11.6	11.5	11.8	11.6	11.6	11.5	11.8	12.9	13.4	12.2	11.9	
Precipitation	PrecAvg	5.9	0.2	0.2	0.2	0.3	0.4	0.6	1.1	1.1	0.8	0.6	0.3	0.2	
	PrecMax	9.1	0.5	0.6	1.0	1.0	1.2	1.7	3.3	3.1	1.9	1.7	0.7	0.6	
	PrecMin	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.2	0.1	0.0	0.0	
	PrecStd	1.4	0.1	0.2	0.2	0.3	0.3	0.4	0.8	0.6	0.4	0.3	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	5.8	14.5	25.1	37.4	63.6	70.8	67.2	51.6	35.8	26.2	16.5	
		MCWB	9.3	5.1	13.4	24.2	35.3	54.8	59.5	57.4	47.5	34.9	25.6	15.7	
	2%	DB	1.1	0.5	8.9	20.8	34.9	56.4	65.2	61.8	47.5	33.4	19.5	9.1	
		MCWB	0.3	-0.3	8.1	19.8	33.5	50.0	55.9	54.9	45.0	32.7	18.8	8.4	
	5%	DB	-3.9	-4.3	3.6	16.8	32.9	51.5	61.8	57.4	44.8	31.5	15.8	4.5	
		MCWB	-4.7	-4.9	2.7	16.1	31.9	46.4	53.8	51.8	42.7	30.8	15.2	3.9	
	10%	DB	-8.2	-10.1	-2.1	11.6	30.6	47.3	58.8	53.9	42.4	29.4	11.5	-1.5	
		MCWB	-8.7	-10.7	-2.9	10.8	29.4	43.3	52.0	49.7	40.8	28.6	10.8	-2.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	5.2	13.7	24.3	35.9	55.3	60.3	58.7	48.0	35.3	25.8	15.9
			MCDB	10.1	5.9	14.7	24.9	36.9	63.2	68.7	65.1	50.6	35.6	26.2	16.4
2%		WB	0.4	-0.2	8.2	19.8	33.7	50.4	56.7	55.2	45.3	32.9	18.9	8.5	
		MCDB	1.0	0.5	9.0	20.7	34.6	56.0	63.8	61.1	47.1	33.3	19.3	9.0	
5%		WB	-4.6	-5.0	2.7	16.0	32.0	46.5	54.5	52.5	43.0	30.9	15.2	3.8	
		MCDB	-4.0	-4.4	3.5	16.8	32.9	51.2	60.7	56.7	44.5	31.4	15.8	4.4	
10%		WB	-8.7	-10.7	-2.8	10.8	29.3	43.5	52.4	50.1	41.0	28.6	10.9	-2.2	
		MCDB	-8.2	-10.2	-2.0	11.5	30.5	47.1	58.1	53.3	42.3	29.3	11.5	-1.6	
Mean Daily Temperature Range		5% DB	MDBR	11.2	11.6	12.8	14.6	12.0	9.8	12.8	10.1	7.0	9.6	11.9	11.4
			MCDBR	16.5	16.9	15.6	14.6	9.9	17.4	18.6	16.3	10.5	7.9	13.4	16.1
	MCWBR		16.3	16.7	15.4	14.6	9.4	12.1	11.5	10.3	8.1	8.2	13.4	16.1	
	5% WB	MCDBR	16.6	17.1	15.5	14.6	9.7	17.3	17.9	15.4	10.0	8.0	13.3	16.0	
		MCWBR	16.4	16.9	15.4	14.6	9.4	12.3	12.0	10.5	8.0	8.4	13.4	16.0	
Clear-Sky Solar Irradiance	taub	0.146	0.210	0.243	0.292	0.310	0.353	0.358	0.337	0.300	0.233	0.161	N/A		
	taud	1.898	2.043	2.221	2.130	2.172	2.284	2.374	2.461	2.471	2.140	1.888	N/A		
	Ebn at Noon	24	180	254	268	276	266	261	254	234	165	13	0		
	Edh at Noon	1	14	23	34	37	35	31	25	19	12	1	0		
All-Sky Solar Radiation	RadAvg	6	132	626	1385	1987	1877	1639	1061	510	157	19	0		
	RadStd	1	10	32	59	83	125	162	86	53	27	2	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page